# Telecom Customer Intelligence Platform

How the engine decides: numbers, formulas, and decision logic (no code)

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The system takes a customer's profile and runs it through a fixed sequence of calculations to reach a business decision. The AI language model never invents or calculates any number - every figure below comes from this deterministic pipeline. This document summarizes only the math and the decision rules.

## 1. Churn Prediction

A trained SVM model outputs a churn probability (0 to 1) for the customer. A fixed decision threshold, baked into the trained model file, converts that probability into a binary decision:

```
Decision threshold = 0.64
```

If churn probability ≥ 0.64 → decision = CHURN. Otherwise → NO CHURN.

### Risk Band (for display / segmentation only)

| Churn probability | Risk Band |
|---|---|
| < 0.25 | Low Risk |
| 0.25 - 0.50 | Medium Risk |
| 0.50 - 0.75 | High Risk |
| ≥ 0.75 | Critical Risk |

## 2. Financial Value of the Customer

These are scenario assumptions, not measured accounting data. Two constants come from the trained model file:

```
Value horizon = 12 months Margin rate = 40%
```

### Estimated Customer Value (historical proxy)

```
EstimatedCustomerValue = MonthlyCharges × tenure (months)
```

Used only to classify the customer as High Value or Low Value (cutoff = 1,376.85). At or above the cutoff → High Value.

### Estimated Future Revenue & Margin (12-month scenario)

```
EstimatedFutureRevenue = MonthlyCharges × 12 EstimatedFutureMargin =
EstimatedFutureRevenue × 0.40
```

### Expected Churn Loss

```
ExpectedChurnLoss = ChurnProbability × EstimatedFutureMargin
```

This is an expected value, not a prediction of actual loss - it blends how likely the customer is to leave with how much margin they would take with them over the next year. It is the key input that drives which retention campaign, if any, is worth spending money on (Section 4).

## 3. Segmentation

Every customer gets two independent labels: a coarse Value x Risk segment, and a more actionable Behavioral segment.

### 3.1 Value x Risk (4 segments)

```
Value Band: High Value if EstimatedCustomerValue ≥ 1,376.85, else Low Value. Risk Level
for this split only: High Risk if churn probability ≥ 0.60 (a separate cutoff from the
4-band Risk Band above - used only here).
```

Combining the two gives: High Value+High Risk, High Value+Low Risk, Low Value+High Risk, Low Value+Low Risk.

### 3.2 Behavioral segment (9 possible labels)

Evaluated top to bottom; the first rule that matches wins (a customer only ever gets one label, even if several rules would technically apply):

| # | Segment | Rule (all conditions must hold) |
|---|---|---|
| 1 | High-Value At-Risk | High Value AND Risk Band is High or Critical |
| 2 | New & Fragile | tenure ≤ 12 months AND Risk Band is High or Critical |
| 3 | Service Gap | Missing Tech Support or Online Security AND Risk Band is High or Critical |
| 4 | Price-Sensitive | Month-to-month contract AND MonthlyCharges > 70.35 AND Risk Band is Medium/High/Critical |
| 5 | Loyal Underpenetrated | tenure ≥ 24 months AND Risk Band is Low/Medium AND missing 2+ services |
| 6 | High-Value Loyal | High Value AND tenure ≥ 24 months AND Risk Band is Low/Medium |
| 7 | Cross-Sell Opportunity | Risk Band is Low/Medium AND missing 1+ services |
| 8 | Low-Value High-Risk | Low Value AND Risk Band is High or Critical |
| 9 | Stable Low-Value | Low Value AND Risk Band is Low or Medium |

"Service Recovery" is deliberately not implemented - it would need real complaint/ticket data, which this dataset does not have.

### 3.3 Churn drivers (why the customer is at risk)

A rule-of-thumb list, not a true model explanation (SHAP). Up to 5 of these are shown, in this order, whichever apply: month-to-month contract, no Tech Support, no Online Security, tenure ≤ 12 months, Fiber optic internet (higher cost), Electronic check payment, paperless billing.

## 4. Retention Campaign Selection (customer predicted to CHURN)

Seven possible campaigns exist, each with its own eligibility condition, cost, and a base success-rate assumption:

| Campaign | Eligible if | Cost | Base success rate |
|---|---|---|---|
| Contract Loyalty Upgrade | Contract = Month-to-month | $30 | 45% |
| Free Tech Support Trial | TechSupport = No | $15 | 40% |
| Free Online Security Trial | OnlineSecurity = No | $15 | 38% |
| Fiber Optic Speed Upgrade | InternetService = Fiber optic | $25 | 42% |
| Auto-Pay & Digital Discount | PaymentMethod = Electronic check | $10 | 32% |
| Early-Tenure Loyalty Discount | tenure ≤ 12 months | $20 | 35% |
| High-Touch Retention Call | Always eligible (fallback) | $40 | 40% |

### 4.1 Success rate is adjusted per customer

The base rate above is the same for everyone. To avoid always recommending the same highest-base-rate campaign, the engine adds small, capped bonuses specific to this customer:

```
adjusted rate = base rate + 12% if the campaign directly addresses one of this customer's churn drivers + 5% if the campaign's target list includes this customer's behavioral segment (result is capped between 5% and 80%)
```

Example: "Contract Loyalty Upgrade" (base 45%) becomes 62% for a month-to-month, Price-Sensitive customer (both bonuses apply), but stays at 45-57% for another eligible customer whose segment or drivers don't match.

### 4.2 Ranking: Expected Net Benefit, not success rate alone

```
expected value preserved = adjusted success rate × ExpectedChurnLoss expected net benefit = expected value preserved − campaign cost expected ROI = expected net benefit / campaign cost
```

The engine picks the campaign with the highest Expected Net Benefit among the ones this customer is eligible for - not the one with the highest success rate. A cheap campaign with a modest success rate can beat an expensive one with a high rate if its net benefit is larger.

If every eligible campaign has a Net Benefit ≤ 0 (i.e. none is worth its cost for this customer), the recommendation becomes "Low-Cost Monitoring" (no paid campaign) instead of forcing an uneconomical offer.

## 5. Upsell / Growth Scoring (customer predicted NO CHURN)

For each of 6 optional services the customer does not already have (Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies), an opportunity score is built:

```
score = 20 (base) + 20 if customer is High Value + 20 if churn probability < 0.30 + 15 if tenure ≥ 24 months + 15 if contract is One year or Two year

annual revenue = monthly price of the service × 12 annual margin = annual revenue × 40%
```

The top 3 services are kept, sorted first by score, then by annual margin. A service the customer already has is never offered again.

## 6. Putting It Together: the Decision Flow

Every customer goes through the same fixed sequence:

1. Data Quality check - flags "LIMITED" if any core field is missing, without inventing data.
2. Churn Prediction (Section 1) - probability, CHURN/NO CHURN decision, Risk Band.
3. Financial calculation (Section 2) - value, future revenue/margin, expected churn loss.
4. Segmentation (Section 3) - Value x Risk segment + Behavioral segment + churn drivers.
5. Branch on decision:
   - CHURN → pick best retention campaign (Section 4).
   - NO CHURN → pick best upsell opportunity (Section 5).

The result of that branch is summarized as one final label - "Next Best Action" - such as Contract upgrade, Service trial, High-touch retention, Low-cost retention, Cross-sell, Upsell, or No action. This label adds no new numbers; it just names what the campaign/upsell engine already decided.

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Note: all dollar figures, rates, and thresholds above (margin rate, value horizon, campaign costs/success rates, decision threshold, value cutoff) are scenario assumptions stored in the trained model file and the campaign catalog - not measured company financial data.